Fiscal Year:	FY 2007	**Task Last Updated:**	FY 06/28/2007
PI Name:	Angelaki, Dora Ph.D.		
Project Title:	Multisensory Interactions to Discriminate Gravity From Translational Acceleration		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Organismal biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology		
Space Biology Cross-Element Discipline:	(1) Developmental Biology (2) Neurobiology		
Space Biology Special Category:	(1) Cell Culture (2) Translational (Countermeasure) Potential		
PI Email:	angelaki@pcg.wustl.edu	**Fax:**	FY 314-747-5069
PI Organization Type:	UNIVERSITY	**Phone:**	314-747-5529
Organization Name:	Washington University Medical School		
PI Address 1:	660 South Euclid Avenue, Box 8108		
PI Address 2:			
PI Web Page:			
City:	Saint Louis	**State:**	MO
Zip Code:	63110-1093	**Congressional District:**	1
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Space Biology 03-OBPR-03
Start Date:	07/01/2004	**End Date:**	06/30/2007
No. of Post Docs:	3	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Dickman, J David (Washington University Medical School)		
Grant/Contract No.:	NNA04CC77G		
Performance Goal No.:			
Performance Goal Text:			
	Throughout the history of the manned space flight program, microgravity has produced vestibular-related symptoms that result in personal discomfort and a loss in crew performance. The symptoms subside within several days of microgravity exposure, suggesting that interactive multisensory mechanisms may be responsible for the initiation of symptoms and the subsequent adaptation. Specifically, the processing and detection of gravity and translational accelerations is one of the most fundamental issues for understanding vestibular sensorimotor control in normal gravity, as well as in altered gravity environments. In order to better understand the nature of multisensory signals in detecting gravity, we will characterize how gravity and translational accelerations are encoded by vestibular neurons in the cerebellum of alert primates. Unless these multi-sensory interactions are better understood in normal gravity, our understanding of neurovestibular adaptation to altered gravity and the development of relevant countermeasures remains largely		

Task Description:	incomplete. The proposed studies aim at filling an important gap in existing knowledge through state-of-the art electrophysiological and computational techniques. The proposed studies aim at filling an important gap in existing knowledge through state-of-the art electrophysiological and computational techniques. Aim I: Investigate the vestibular convergence patterns in the rFN and NVU: We propose to use combinations of three-dimensional (3D) rotations and translations in order to characterize the otolith/semicircular canal convergence patterns in the rFN and NVU. Specifically we propose: (1) To examine if there are populations of rFN and NVU neurons with properties similar to those found in the VN (Canal-Only, Otolith+Canal and Otolith-Only cells). (2) To quantify differences and/or similarities in the 3D distributions of rotational maximum sensitivity vectors using earth-vertical axis (EVA) rotation (canal only stimulation), as well as in the spatial distributions of translational sensitivity vectors (otolith only stimulation) between convergent and non-convergent cells. (3) To compare EVA rotation and translational responses with cell firing rate modulation during earth-horizontal axis (EHA) rotations in order to examine the pattern of otolith/canal convergence in the rFN and NVU. Our working hypothesis is that the rFN and NVU circuitry is specific to the processing of linear acceleration. Thus, unlike the VN where a quarter of the sampled neuronal population was identified as Canal-Only cells, we hypothesize that we will encounter very few, if any, Canal-Only neurons in the rFN and NVU. Aim II: Test if rFN and NVU neurons discriminate dynamic tilts relative to gravity from translational components of linear acceleration: Although primary otolith afferents respond equally to translational acceleration and dynamic tilts relative to gravity, central vestibular neurons could selectively encode the translational component of the motion. We will quantitatively test this hypothesis using protocols with combinations of translational and tilt motion stimuli. We hypothesize that a subpopulation of rFN and NVU neurons, including all neurons antidromically identified as projecting to the VN, will selectively respond to translational head motion and ignore changes in head orientation relative to gravity. Aim III: Investigate the coordinate system (head vs. body-fixed) for encoding translational motion by rFN and NVU neurons: Primary otolith afferents encode linear acceleration in a head-fixed coordinate system. However, spatial orientation, postural control and self-motion awareness might require that there exist central neurons encoding translational motion of the body in space. In these studies we will investigate the influence of static neck inputs in the processing the otolith information by systematically varying both the direction of sled motion relative to the body and the static orientation of the head relative to the trunk. Because both coordinate systems might be important for downstream processing, we hypothesize that translation-coding rFN and NVU neurons that project to the VN include both head-, as well as body-fixed coordinate systems. Throughout the history of the manned space flight program, microgravity has produced vestibular-related symptoms that result in personal discomfort and a loss in crew performance. The symptoms subside within several days of microgravity exposure, suggesting that interactive multisensory mechanisms may be responsible for the initiation of symptoms and the subsequent adaptation. Specifically, the processing and detection of gravity and translational accelerations is one of the most fundamental issues for understanding vestibular sensorimotor control in normal gravity, as well as in altered gravity environments. In order to better understand the nature of multisensory signals in detecting gravity, we will characterize how gravity and translational accelerations are encoded by vestibular neurons in the cerebellum of alert primates. Unless these multi-sensory interactions are better understood in normal gravity, our understanding of neurovestibular adaptation to altered gravity and the development of relevant countermeasures remains largely incomplete. The proposed studies aim at filling an important gap in existing knowledge through state-of-the art electrophysiological and computational techniques.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The proposed studies are also directly relevant to understanding complex processing and multisensory interactions in the brain. In addition to space flight, this knowledge is important for understanding brain function and computational neural networks.
Task Progress:	We have made outstanding progress and we have successfully finished all aims of the grant. The most noticeable contribution is that we have identified neuronal populations that encode a neural estimate of gravitational accelerations and translational motion. This is a huge contribution because, like all linear accelerometers, primary otolith afferents encode net gravitoinertial acceleration and cannot distinguish gravitational from translational accelerations. In these studies we have shown that all Purkinje cells from lobules 9 and 10 of the cerebellar vermis, as well as a subpopulation of neurons in the vestibular and deep cerebellar nuclei carry the appropriate otolith and semicircular canal signals to be able to separate the two components of linear acceleration. This work has been published in high profile journals, like Nature, Neuron and Current Biology. I have provided below the summary for parts of this work. The ability to orient and navigate through the terrestrial environment represents a computational challenge common to all vertebrates. It arises from the fact that the motion sensors in the inner ear, the otolith organs and the semicircular canals, transduce self-motion in an egocentric reference frame. As a result, the vestibular afferent information reaching the brain is inappropriate for coding our own motion and orientation relative to the outside world. Here we show that activity in cerebellar cortical neurons in lobules 9 and 10 of the vermis reflects the critical computations of transforming head-centered vestibular afferent information into earth-referenced self-motion and spatial orientation signals. Unlike vestibular and deep cerebellar nuclei neurons where a mixture of responses was observed, Purkinje cells represent a homogeneous population that encodes inertial motion. They carry the earth-horizontal component of a spatially-transformed and temporally-integrated rotation signal from the semicircular canals, which is critical for computing head attitude, thus isolating inertial linear accelerations during navigation. The most medial of the deep cerebellar nuclei, the fastigial nucleus (FN), receives sensory vestibular information and direct inhibition from the cerebellar vermis. We investigated the signal processing in the primate FN by recording single unit activities during translational motion, rotational motion and eye movements. Firing rate modulation during horizontal plane translation in the absence of eye movements was observed in all non-eye movement-sensitive cells and 26% of the pursuit-eye movement sensitive neurons in the caudal FN. Many non-eye-movement-sensitive cells recorded in the rostral FN of three fascicularis monkeys exhibited convergence of signals from both the otolith organs and the semicircular canals. At low frequencies of translation, the majority of these rostral FN cells changed their firing rates in phase with head velocity, rather than linear acceleration. As frequency increased, FN vestibular neurons exhibited a wide range of response dynamics, with most cells being characterized by increasing phase leads as a function of frequency. Unlike cells in the vestibular nuclei, none of the rostral FN cells responded to rotational motion alone,

	without simultaneously exhibiting sensitivity to translational motion. Modulation during earth-horizontal axis rotation was observed in more than half (77%) of the neurons, although with smaller gains than during translation. In contrast, only 47% of the cells changed their firing rates during earth-vertical axis rotations in the absence of a dynamic linear acceleration stimulus. These response properties suggest that the rostral FN represents a main processing center of otolith-driven information for inertial motion detection and spatial orientation.
Bibliography Type:	Description: (Last Updated: 04/21/2017)
Articles in Peer-reviewed Journals	Shaikh AG, Meng H, Angelaki DE. "Multiple reference frames for motion in the primate cerebellum." J Neurosci. 2004 May 12;24(19):4491-7. PMID: 15140919 , May-2004
Articles in Peer-reviewed Journals	Green AM, Angelaki DE. "An integrative neural network for detecting inertial motion and head orientation." J Neurophysiol. 2004 Aug;92(2):905-25. Epub 2004 Mar 31. PMID: 15056677 , Aug-2004
Articles in Peer-reviewed Journals	Angelaki DE, Shaikh AG, Green AM, Dickman JD. "Neurons compute internal models of the physical laws of motion." Nature. 2004 Jul 29;430(6999):560-4. PMID: 15282606 , Jul-2004
Articles in Peer-reviewed Journals	Shaikh AG, Ghasia FF, Dickman JD, Angelaki DE. "Properties of cerebellar fastigial neurons during translation, rotation and eye movements." J Neurophysiol. 2005 Feb;93(2):853-63. Epub 2004 Sep 15. PMID: 15371498 , Feb-2005
Articles in Peer-reviewed Journals	Green AM, Shaikh AG, Angelaki DE. "Sensory vestibular contributions to constructing internal models of self-motion." J Neural Eng. 2005 Sep;2(3):S164-79. Review. PMID: 16135882 , Sep-2005
Articles in Peer-reviewed Journals	Shaikh AG, Green AM, Ghasia FF, Newlands SD, Dickman JD, Angelaki DE. "Sensory convergence solves a motion ambiguity problem." Curr Biol. 2005 Sep 20;15(18):1657-62. PMID: 16169488 , Sep-2005
Articles in Peer-reviewed Journals	Yakusheva TA, Shaikh AG, Green AM, Blazquez PM, Dickman JD, Angelaki DE. "Purkinje cells in the posterior cerebellar vermis encode motion in an inertial reference frame." Neuron. 2007 Jun 21;54(6):973-85. PMID: 17582336 , Jun-2007
Articles in Peer-reviewed Journals	Angelaki DE, Cullen KE. "Vestibular system: The many facets of a multimodal sense." Annu Rev Neurosci. 2008;31:125-50 [Epub ahead of print] . Review. PMID: 18338968 ; Originally reported in 2007 as [Epub ahead of print] in March 2008. , Mar-2008